

The background is a dark blue gradient. On the left, there is a large, semi-transparent circular image of a circuit board. Overlaid on this and the background are several geometric shapes: a large blue triangle pointing downwards and a green triangle pointing upwards, both with black outlines. In the top right corner, there is a 3D perspective view of a circuit board layout.

Car Price Category Prediction using Decision Trees

David Forero, Robert Castanon



Introduction

The value of cars can be unpredictable to an untrained eye. Is it the brand name? Is it the car's performance? By using this dataset we will be able to predict the price class of any car whether they fall under affordable for everyone ($< \$15,000$), affordable for middle-high class ($\$15,000 - \$45,000$), affordable for elite class ($> \$45,000$) for users who quickly want to find out if a car is within their price range. Using a decision tree classifier, our model will accurately predict a car's price category given a set of features input by the user.



Dataset

Source: Kaggle - Cars Datasets (2025) by Abdul Malik
(<https://www.kaggle.com/datasets/abdulmalik1518/cars-datasets-2025>)

Type: Structured

Instances: 1201 values

Features: Car Company Names, Car Models, Engine Types, CC/Battery Capacity, Horsepower (HP), Top Speed, 0-100 km/h Performance, Price (in USD), Fuel Type, Seating Capacity, Torque

Format: .csv, encoding=cp1252

Classes for Classification: Car prices will be categorized into the following:

Budget (<\$15,000), Medium (\$15,000 - \$45,000), Luxury (>\$45,000)

Dollar signs and commas removed from prices. Prices rounded to nearest 1000.



Classification Algorithms

Decision Tree Classifier from scikit-learn

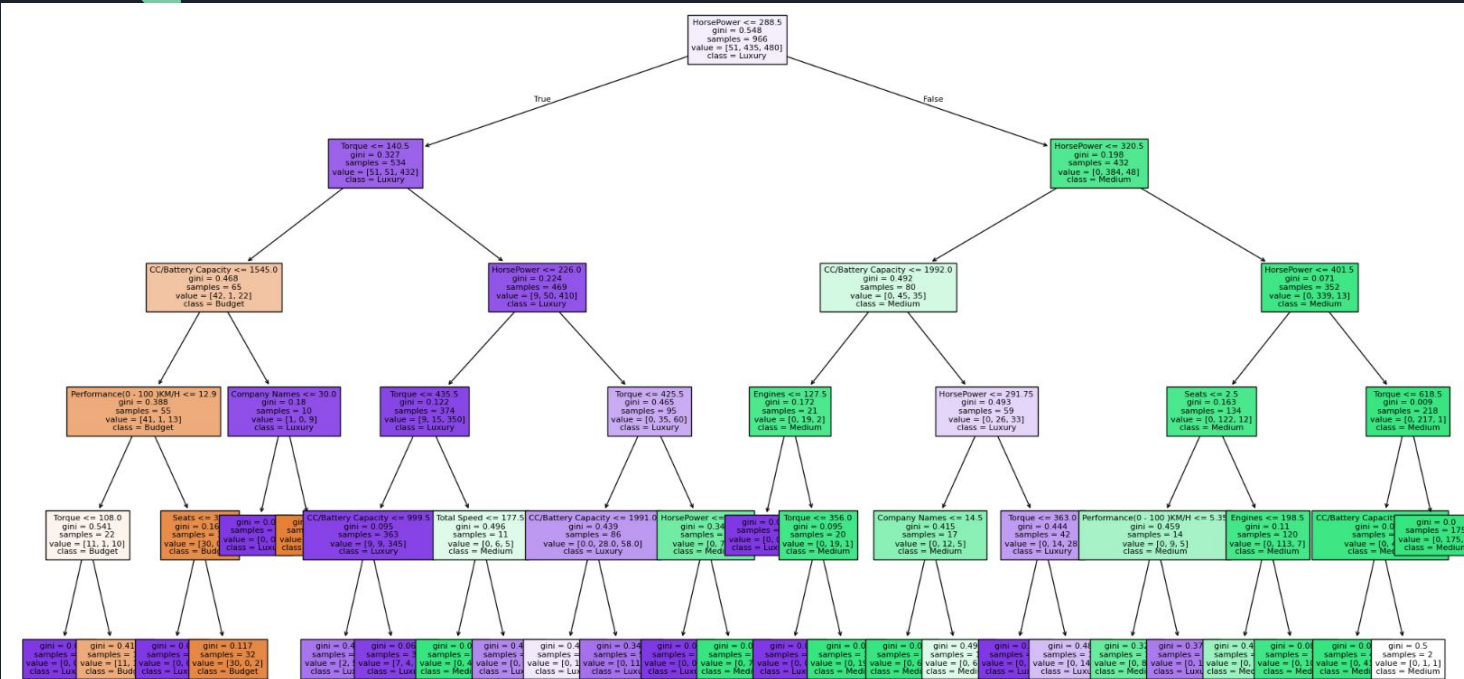
Used an 80/20 train-test split as shown in lecture

Depth = 5; Criterion = gini; random state = 42

Label Encoder used



Results



Model Accuracy:
85%

Model accepts user
input and returns
cars similar to the
requested.

Model displays all
cars within
requested/given
category.



Discussion & Analysis

Performed well with 85% accuracy

Had problems using dataset

Limitations included: ordering of categorical variables and large space between budget and luxury (15,000 - 45,000)

Model focused on Horsepower, Torque, and Engine Capacity for price classification

Could improve using One-Hot Encoding, more pruning, and algorithms like Random Forest



Teamwork and Task Division

David Forero -

Coding, Debugging, User Input, Dataset Acquisition, Slideshow Preparation

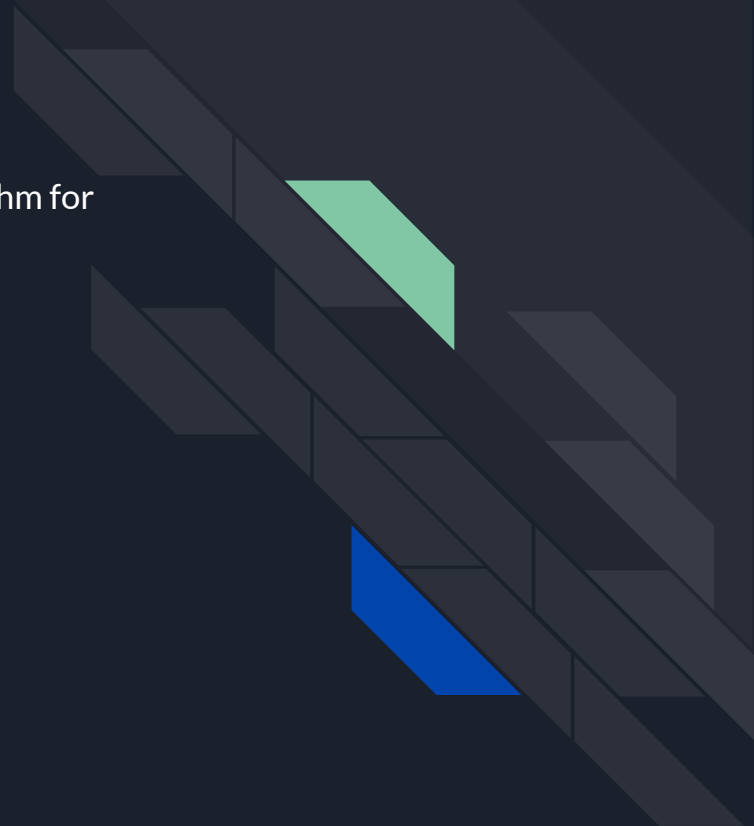
Robert Castanon -

Coding, Debugging, Decision Tree Visualization, Dataset Formatting, Slideshow Preparation



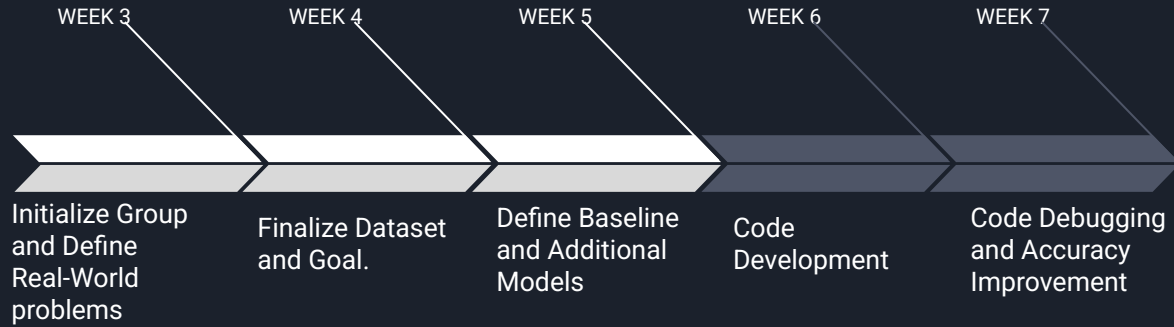
Lessons Learned

- Importance of finding and implementing the correct algorithm for the goal trying to be achieved.
- The uses of sklearn Library
- Importance on visual data and graphs for user clarity.





Project timeline





Thank you!